

CE players. The ORB core allows us to build and integrate rich graphics, feature-rich media and convergence applications into next-generation devices," says Borker.

Low bandwidth demands and cost-efficiency. TveeBoX uses NextBIT Computing's patented Modya client-server architecture, which results in huge savings and scalability to the operation of a video-on-demand network.

"The patented Modya video-on-demand server architecture and middleware technology features a mechanism to store, encrypt and deliver secured media over an Internet connection. The mechanism of storage and transfer employed in Modya results in reduction of bandwidth requirements and bandwidth costs to 3 per cent of what would traditionally be required. This is achieved through a content encoding mechanism that we have devised. This patented mechanism of encoding also features an integrated DRM, allowing content owners to deliver content to the end user at no additional content DRM costs," says Borker.

Further, the use of Linux operating system and GNU tools helps keep the software costs in check. As a result, a basic media adaptor STB with cloud PC interface costs around \$160 and the more advanced DVB-C PVR box with cloud PC and video phone costs around \$250. The device is also power-efficient, consuming a peak power of 10 watts and minimum of 4 watts. Thus, TveeBoX is light on the pocket.

Growing user experience

TveeBoX ensures there is no dent in the user experience. For instance, there is no erosion in the quality of user-generated content of websites like YouTube



The NextBIT team

fitted to the TV resolution.

"Due to the higher resolution of TV, videos (like YouTube videos) may lose quality in the process of scaling to fit the TV resolution. We developed efficient post-processing algorithms that allow better display of content rather than the lousy display otherwise observed. The beauty of the video remains or gets enhanced due to the post-processing techniques we employ," vouches Borker.

Further, NextBIT is working on an application store that will allow the TveeBoX users to download new applications released for the TveeBoX platform. It plans to release widgets and application interfaces to the periodic top 50 Internet portals.

Path wasn't easy though

Developing the TveeBoX from ground up—including hardware design, testing, validation, operating system porting, driver development and bug fixing, development of multimedia and IPTV middleware, user interfaces and applications—wasn't a cakewalk. A development team of over 18 engineers spent more than three years to build this feature-rich and powerful convergent STB platform. They surpassed multiple challenges *en route*, like the lack of

a manufacturing and sourcing ecosystem and support from processor companies.

"It is quite a financial drain getting prototypes developed in India. Further, we were not getting support from processor companies. Today, we are forced to look towards China for mass manufacturing due to reasons of cost-effectiveness. Thereafter, keeping the technology minds working at NextBIT aspired was also a challenge," recalls Borker.

The road ahead!

NextBIT has begun mass manufacturing of two models. It is working directly with operators and, over the next 18 months, plans to enter the open STB market for hybrid digital cable and IP services.

Looking ahead, NextBIT plans to introduce the server gaming concept on the TveeBoX. It is also following the developments in the open source community and evaluating how best it can use them to enhance the user experience.

"Over a period of time, when we have substantial user base, we hope to open up our development APIs to allow a more open method of adding user-generated features and applications to our devices," says Borker.

Alongside, it is in talks for funding of another product expected to be launched in August. ●

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